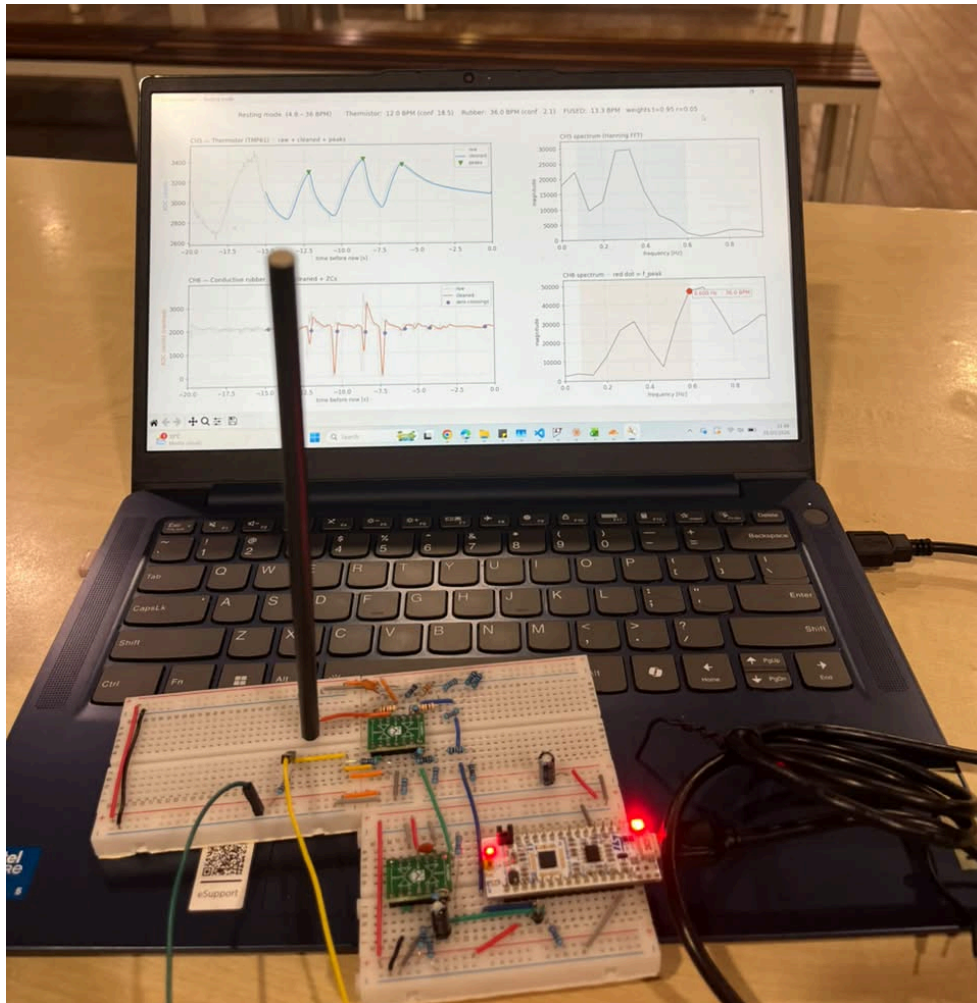


MONASH University

Project 3: Breath Rate Estimator

Sensor Characterisation Report



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1. Sensor and Circuit

1.1 Sensor Selection and Rationale

A TMP61 thermistor and a conductive rubber cord were used as input sensors in combination, as they transduce breathing through fundamentally different physical principles with different failure/constraints. The thermistor, using a straw to blow air into it, detects the temperature swing between cool inhaled air and warm exhaled air, giving a direct measure of airflow but leaving it vulnerable to ambient draughts and head movement. The rubber cord, worn as an elastic band around the abdomen, transduces thoracic expansion through a change in its bulk resistance under strain, making it robust to airflow disturbances, but susceptible to motion artefacts and slow viscoelastic baseline drift. Pairing both sensors ensures a disturbance that defeats one sensor is unlikely to defeat the other, which is the foundation on which the data-fusion stage in Section 4 relies.

1.2 Common Signal Conditioning Topology

Both sensors are resistive, so each channel begins with a fixed-resistor voltage divider that converts the sensor resistance into a voltage equivalent. To prevent the downstream amplifier network from loading the divider and corrupting its ratio, the divider output is fed into an op-amp wired as a unity-gain buffer for impedance matching. The buffered signal then drives a different amplifier whose inverting input is held at a fixed reference voltage; this stage subtracts the standing DC offset and applies channel-specific gain so that the small AC component due to breathing fills as much of the 0 – 3.3 V ADC window as possible. The full schematic and breadboard photograph are provided in [Appendix B](#).

1.3 Thermistor Channel

The thermistor divider uses the 10 k Ω series reference resistor recommended in the project brief, which matches the TMP61's nominal 10 k Ω resistance at 25 °C and places the divider output near mid-rail. With the sensor placed inside a straw, the buffered output was measured at approximately 1.70 V at rest, which is slightly above mid-rail because skin-proximity raises the thermistor temperature above ambient.

To cancel this DC level inside the difference amplifier, a reference divider built from two 2.2 k Ω and two 100 Ω resistors generates

$$V_{ref, therm} = V_{cc} \times \frac{(100||100)\Omega + 2.2 k\Omega}{2.25 k\Omega + 2.2 k\Omega} = 3.3 V \times \frac{2.25 k\Omega}{4.45 k\Omega} = 1.67 V \dots (1)$$

The amplifier is configured for

$$A_v = \frac{R_F}{R_{in}} (V_{therm} - V_{ref, therm}) = \frac{470 k\Omega}{10 k\Omega} (V_{therm} - 1.67) = 47(V_{therm} - 1.67) V/V, \quad V_{therm} \geq 1.70 V \dots (2)$$

so that a typical breath-driven swing of around 30 mV at the thermistor divider expands to roughly 1.4 V at the ADC input. Recorded data confirms this: across a five-cycle resting trial, the CH5 (Pin A0) ADC counts ranged from 531 to 3222, a swing of approximately 66 % of full scale, with no railing at either supply.

1.4 Conductive Rubber Channel

The conductive rubber cord was measured with a multimeter at 980 Ω in its relaxed, unstretched state, which directly motivated the choice of a 1 k Ω series reference resistor for its divider. This places the unstrained operating point at

$$V_{rubber, rest} = V_{cc} \times \frac{1 k\Omega}{980 \Omega + 1 k\Omega} = 3.3 V \times \frac{1 k\Omega}{1.98 k\Omega} = 1.67 V \dots (3)$$

which is close to mid-rail and well matched to a 1 k Ω / 1 k Ω reference divider (1.65 V) on the difference amplifier's inverting input. The gain was deliberately set lower than on the thermistor path,

$$A_{v, rubber} = \frac{R_F}{R_{in}} (V_{rubber} - V_{ref, rubber}) = \frac{1 k\Omega}{100 \Omega} (V_{rubber} - 1.65) = 10(V_{rubber} - 1.65) V/V, \quad V_{rubber} \approx 1.67 V \dots (4)$$

because the rubber's strain-induced resistance change per breath is roughly an order of magnitude larger than the thermistor's, and bench testing showed that a 47 $\times$ gain on this channel saturated the amplifier during deeper inhalations.

A second design issue emerged during testing, where the raw rubber waveform is non-uniform, with irregular peaks in a cycle rather than a single clean inhale-exhale transition per breath. Fed directly into the peak and zero-crossing-based DSP in Section 3.3, these false peaks caused the algorithm to register multiple counts per single breath cycle, inflating the estimated breath rate. To suppress these fast irregularities before they reached the gain stage, the input to the final differential amplifier was filtered by a 10 μ F capacitor and a 1 k Ω resistor at the corner frequency

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 1 k\Omega \times 10 \mu F} = 15.9 Hz \dots (5)$$

which sits comfortably above the breathing band (well under 2 Hz even during heavy exercise) but well below the frequencies of irregular and false peaks that arise within a cycle, so the breath envelope passes unaltered while the higher-frequency irregularities are attenuated. The smoothed rubber signal is then differentiated against the thermistor amplifier output and amplified by 100 $\times$, which drives the stage hard into both supply rails on every breath cycle. The result is a near-square-wave output whose transitions mark the moments of maximum chest expansion and recovery, a deliberate trade-off, since the rubber-channel DSP extracts breath rate from the timing of these transitions via FFT and zero-crossing analysis, and preserving the absolute amplitude of the strain signal is unnecessary. The recorded data confirms the design

intent where CH6 (Pin A1) occupies the full 0 – 4053 ADC range with sharp edges aligned to each breath cycle, in both the resting and exercise recordings.

2. STM32 Acquisition and Data Streaming

The conditioned outputs of the two chains are routed to the STM32’s on-board ADC inputs with the thermistor connected to pin A0 (CH5) and the differential rubber channel linked to pin A1 (CH6). Both channels are sampled at a constant 50 Hz, combined and continuously transmitted to the host over UART as comma-separated text lines of the form “CH5,CH6\r\n”. The firmware begins streaming immediately on reset, while the host side manages trial start and stop using the Python tool outlined in Section 3. The sampling rate of 50Hz sits well above the Nyquist requirement for breathing signals, whose maximum meaningful frequency is approximately 1.5 Hz during exercise conditions, while leaving sufficient headroom for the moving-average and wavelet stages in Section 3, both of which gain advantages from a high sample-to-cycle ratio. The readings sent by the STM32 through the serial channel are shown in [Appendix C, as shown in Figure C.1](#).

3. Digital Signal Processing and Algorithm

The DSP pipeline is designed primarily for noise reduction since both raw sensor channels have significant out-of-band signals that interfere with the breathing signal at the edges: ADC quantisation chatter, mains-frequency pickup on the high-impedance buffer inputs, and motion-induced amplifier hash. Every recorded trial initially goes through a four-stage preprocessing pipeline that is identical for both channels (Section 3.1), subsequently being processed by channel-specific estimators that exploit the very different morphologies of the two waveforms. A peak-counting estimator for the smooth, near-sinusoidal thermistor signal (Section 3.2) and an FFT-driven, envelope-gated zero-crossing estimator for the railed, square-wave-like rubber signal (Section 3.3). All host-side processing is implemented in `breath_analyzer.py` (see Appendix A for the repository).

3.1 Common Preprocessing Pipeline

Figure 3.1.1 displays the entire DSP pipeline in a block diagram format from start to finish. Both raw ADC channels enter the common preprocessing stage outlined in this section, after which the cleaned signals diverge into two estimator branches whose outputs are recombined at the data-fusion stage in Section 4. The branches were developed separately as the two signal morphologies are too distinct to share a single estimator. The thermistor produces a smooth, near-sinusoidal waveform whose breath count is trivially read from its local maxima, whereas the rubber channel is a heavily saturated near-square wave whose breath rate lives in the interval between its rail-to-rail transitions. Imposing a common estimator on both would force one of them onto an algorithm tuned for the incorrect signal class.

DSP Pipeline Block Diagram

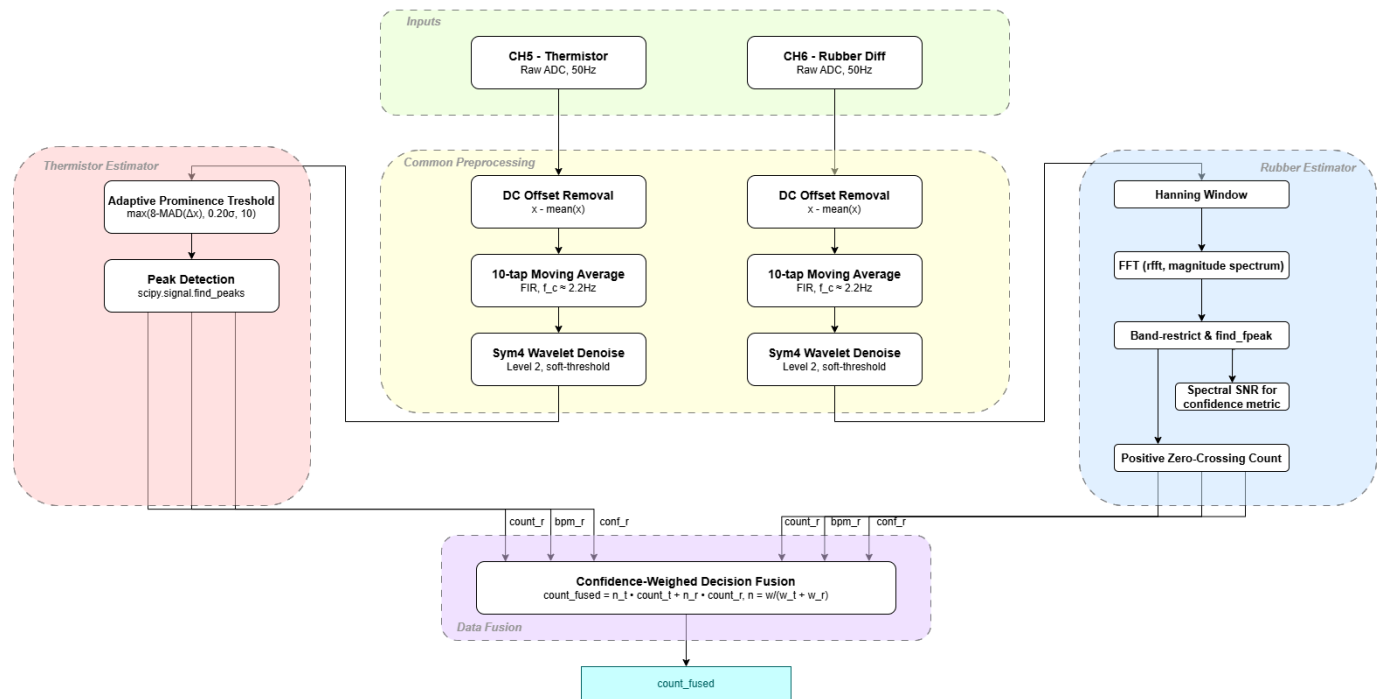


Figure 3.1.1. Block diagram of the full end-to-end DSP Pipeline

The shared preprocessing function performs four operations in a fixed order. DC offset removal, FIR low-pass smoothing, wavelet denoising, and an optional Hanning taper are applied only when the cleaned signal is intended for an FFT. The established sequence demonstrates the reliance among phases; both the moving average and the wavelet denoiser rely on a zero-mean input to behave predictably, and the Hanning taper would distort the time-domain peak and zero-crossing detectors if applied unconditionally.

DC offset removal subtracts the per-trial mean from every sample,

$$x_{ac}[n] = x[n] - \frac{1}{N} \sum_{n=0}^{N-1} x[n] \dots (6)$$

This serves two purposes. Initially, it eliminates the persistent ADC offset of approximately 2050 counts to ensure that the following FFTs aren't overwhelmed by a significant DC bin, which would hinder the rubber-channel f_{peak} search in Section 3.3 from settling on the breathing frequency. Secondly, it transforms the peak-prominence threshold in Section 3.2 into a noise-floor-relative measure instead of an absolute one, meaning the thermistor estimator does not require recalibration when ambient temperature alters the baseline.

A 10-tap moving-average FIR low-pass is then applied via convolution with a uniform kernel $h[n] = 1/10$ for $n = [0, 9]$. The -3 dB cutoff of a length-N moving average at sample rate f_s is approximately

$$f_c \approx \frac{0.443f_s}{N} = \frac{0.443 \times 50 \text{ Hz}}{10} \approx 2.2 \text{ Hz} \dots (7)$$

which sits just above the maximum expected breathing frequency under exercise conditions (1.5 Hz, Section 1.4) and well below the 25 Hz Nyquist frequency. The filter therefore preserves the breathing band intact while attenuating ADC quantisation noise and any high-frequency amplifier hash above 3 Hz. A 10-tap length was selected as the smallest order that achieved this corner without significantly altering the peak times of clean test waveforms.

Sym4 wavelet denoising at decomposition level 2 subsequently addresses the residual in-band noise that is unreachable by the moving average. The signal is decomposed into an approximation array and two arrays of detail coefficients. The Symlet-4 wavelet was selected for its nearly symmetric impulse response and four vanishing moments, allowing for smooth reconstruction of slow physiological signals without the ringing artefacts.

Finally, a Hanning window of the form $w[n] = 0.5(1 - \cos(\frac{2\pi n}{N-1}))$ is utilised exclusively when the cleaned signal is intended for one of the FFT-based stages discussed in Section 3.3, multiplying the signal element-wise so that its edges taper to zero and the spectral leakage of the finite-length DFT is suppressed. Applying the window unconditionally would attenuate the leading and trailing breath cycles in the time domain, biasing both the peak counter and the zero-crossing counter downward. The preprocessing function therefore exposes this stage as an opt-in argument that is used only by the spectral analysis paths. The raw, moving-averaged, and wavelet-denoised stages of the CH5 thermistor signal across a five-cycle resting trial are shown in [Appendix D, Figure D.1](#).

3.2 Thermistor Estimator: Adaptive-Prominence Peak Counting

Following the common preprocessing pipeline, the cleaned thermistor signal appears as a smooth, nearly sinusoidal waveform in which each breath cycle aligns exactly with one local maximum, which is the temperature crest of an exhalation. The estimator therefore reduces to a peak-counting problem solved by `scipy.signal.find_peaks` with two configured parameters, a minimum inter-peak distance and a minimum prominence threshold, selected with clear design reasons instead of being manually adjusted.

The minimum inter-peak distance is taken from the mode parameters and converted to a sample count of $peak_min_dist \times f_s$. In rest mode, this is $2.0 \text{ s} \times 50 \text{ Hz} = 100$ samples, which caps detection at a ceiling of 30 BPM, while in exercise mode, it is $0.8 \text{ s} \times 50 \text{ Hz} = 40$ samples, which caps detection at 75 BPM. Both ceilings are significantly above the physiologically expected upper bound for the relevant condition (resting subjects rarely exceed 20 BPM, exercising subjects rarely exceed 50 BPM), meaning the filter cannot stop an actual breath from being recorded, but it does prevent the algorithm from detecting two peaks within a single cycle when a minor noise spike overlaps with the main lobe.

The minimum prominence threshold eliminates noise-floor fluctuations without discarding genuine shallow breaths. A fixed value will fail in both directions, as tuning for the resting signal misses the smaller amplitudes seen during exercise, while tuning for exercise it lets enough rest-state noise through to inflate the count. The threshold must be adjusted with the signal, which requires an estimate of the noise floor that is not itself corrupted by the breath. The implementation obtains this by working from the first difference of the cleaned signal, Δx , because consecutive samples of a slow breathing waveform are nearly equal; Δx discards almost all breath-amplitude information and retains mainly residual high-frequency noise. Its median absolute deviation (MAD), a robust, outlier-resistant analogue of the standard deviation, is thus a clean noise-only estimate. Taking the MAD of the signal itself would not work: for a sinusoid of amplitude A , the MAD is $\approx A/\sqrt{2} \approx 0.71A$, so it would track the breath swing rather than the noise, which is exactly what an adaptive threshold must avoid.

The threshold combines three terms, each guarding a different failure mode:

$$prominence = \max(8 \cdot MAD(\Delta x), 0.2\sigma_x, 10) \dots (8)$$

The first term, $8 \cdot MAD(\Delta x)$, is the dominant, adaptive component, where the $8 \times$ multiplier was selected based on experience as the smallest factor that minimises false peaks across the test recordings while retaining real ones. The second term, $0.20 \cdot \sigma_x$, is a floor proportional to overall signal amplitude, preventing the threshold from collapsing toward zero if the noise estimate is anomalously small. The third, a hard floor of 10 ADC counts, covers pathological inputs where both adaptive terms degenerate. Together, these let a single expression hold across the rest and exercise regimes that defeat any fixed value.

The estimator outputs three quantities consumed by the data-fusion stage in Section 4: the peak count `count_t`, the resulting breath rate `bpm_t = count_t \times 60 / T`, and a confidence metric.

$$conf_t = \frac{prominence_{peaks}}{8 \cdot MAD(\Delta x)} \dots (9)$$

which expresses the mean detected-peak prominence as a multiple of the noise-floor estimate. A `conf_t` near 1 indicates peaks barely clearing the threshold (untrustworthy), while a `conf_t` of 10 or higher indicates peaks an order of magnitude above the noise floor (highly trustworthy). This scalar is the quantity the fusion stage in Section 4 uses to weigh the

thermistor branch against the rubber branch. The cleaned CH5 thermistor signal across a five-cycle resting trial is shown in [Appendix D, Figure D.2](#).

3.3 Rubber Estimator: Zero-Crossing Counting

After the shared preprocessing of Section 3.1, the cleaned rubber signal is a zero-mean near-square wave in which each breath creates precisely one positive-going crossing of the baseline. The estimator identifies these crossings by locating the sample indices n at which $\text{sign}(y[n+1]) > \text{sign}(y[n])$, then applies the same minimum-distance constraint used in the thermistor branch; a crossing is rejected if it falls within $\text{peak_min_dist} \times f_s$ samples of the previously accepted one (100 samples in rest mode, 40 in exercise mode). This constraint is necessary because the wavelet denoising in Section 3.1 cannot perfectly eliminate every small zero-crossing minor disruption in the transition regions of the saturated waveform, and the minimum-distance filter functions as a refractory period analogous to the lecture's framing of an FIR memory window to prevent a single breath from being counted two or three times if its transition edge contains residual jitter.

A Hanning-windowed FFT of the cleaned signal is computed for two main purposes. The first is the live-display spectrum panel, which helps the operator to visually verify that the dominant spectral energy lies within the expected band for the selected mode. The second is the rubber-branch confidence metric, which is computed as the in-band spectral signal-to-noise ratio,

$$\text{conf}_r = \frac{\max(f \in [f_{\min}, f_{\max}]) \cdot |X(f)|}{|X(f)|_{f \in [f_{\min}, f_{\max}]}} \dots (10)$$

with the mode-dependent band edges of Table 3.3.1. A high conf_r (values near 10 or above) indicates a sharp, well-defined spectral peak corresponding to steady periodic breathing, whereas a low conf_r (near 1) indicates a flat in-band spectrum consistent with irregular breathing, motion artefacts, or sensor instability. The Hanning window was selected over the Hamming window because its complete tapering to zero at both edges provides a slightly narrower main-lobe width and therefore better frequency discrimination, which is more important for an SNR-based confidence measure than the slightly lower side-lobe rejection of Hamming.

Table 3.3.1: Mode-dependent parameters used in both estimator branches.

Mode	FFT Band	Equivalent (BPM)	Min-distance (s)	Max detectable (BPM)
Resting	0.08 – 0.60	4.8 – 36	2.0	30
Exercising	0.30 – 1.50	18 – 90	0.8	75

4. Data Fusion

4.1 Fusion Approach and Design Rationale

The two estimator branches utilise completely independent physical principles and exhibit distinct failure modes. The thermistor branch is vulnerable to ambient airflow disturbances and head movements that displace the sensor from the respiratory path. On the other hand, the rubber channel is susceptible to chest-strap slippage and postural shifts that distort the expansion strain envelope. Hence, the fusion stage has two objectives: to improve on the worse single-sensor baseline by suppressing its errors when the better sensor is healthy, and to provide graceful degradation by reweighting toward the surviving channel when one sensor fails. This is a stronger requirement than a fixed-weight average can meet and a more conservative one than a hard switch, which motivates the two architectural choices that follow.

- The first was to perform fusion at the decision level (combining the BPM and count outputs of the two estimators) rather than at the signal level (combining the raw or cleaned time-series before estimation). Signal-level fusion would require aligning and normalising two waveforms with completely different morphologies, namely the smooth near-sinusoid from the thermistor and a saturated near-square wave from the rubber cord. Doing this would wipe out the custom algorithmic choices that each estimator relies on to do its job. Decision-level fusion preserves these specialisations and confines each branch's failure to its own scalar outputs, where the fusion stage can detect and down-weight it via the confidence metric.
- The second choice was to use continuous confidence-weighting rather than hard switching between sensors or a fixed-weight average. A fixed-weight average would typically respond uniformly to both healthy and degraded conditions. In contrast, a hard threshold switch would be slightly aggressive, discarding useful partial data the moment a sensor becomes marginally degraded. A continuous weighing scheme offers the best of both behaviours because when both sensors are performing well, they both contribute based on their reliability, and when one sensor degrades, its weight shrinks smoothly toward zero without vanishing.

Measured against these objectives, the fusion stage drops rubber-only MSE from 0.44 to 0.00 cycles² at rest and from 1.48 to 0.16 at exercise (Tables 5.2.1, 5.2.2), and Section 4.3 demonstrates the weight rebalancing from $n_t = 0.98$ at rest to 0.42 post-exercise. The fused output is structurally bounded above by the thermistor when both sensors are healthy, a limitation examined in Sections 5.4 and 5.5.

4.2 Confidence-Weighted Decision Fusion

The combined outputs are convex combinations of the two branch estimates with combination coefficients derived from the per-channel confidence scalars defined in Sections 3.2 and 3.3. Both metrics are naturally non-negative and share an intuitive scale where values nearing “1” indicate an unreliable estimate and values of 10 or higher signify high reliability. Raw weights are established by clamping each confidence scalar to a small positive floor ($\varepsilon = 10^{-5}$) to prevent exact zeros, and applying a fixed multiplier to the rubber channel (k_r).

$$w_t = \max(\text{conf}_t, \varepsilon) \quad w_r = k_r \cdot \max(\text{conf}_r, \varepsilon) \dots (11)$$

The scaling factor $k_r = 0.5$ was determined empirically. Across testing regimes spanning rest and exercise conditions, the thermistor consistently produced breath counts within ± 1 cycle of the manual ground truth, whereas the rubber loop exhibited wider tracking variances. If unadjusted ($k_r = 1.0$), periods of high rubber confidence caused by localised tightening of the FFT spectrum would pull the overall estimate toward the less accurate baseline, which would otherwise be a degrading performance. Therefore, the chosen coefficient preserves the supplementary value of the rubber channel while safeguarding the accuracy of the final output. Normalising these raw weights yields the final fusion coefficients:

$$n_t = \frac{w_t}{w_t + w_r}, \quad n_r = \frac{w_r}{w_t + w_r}, \quad n_t + n_r = 1 \dots (12)$$

As well as the fused breath rate and breath count:

$$BPM_{fused} = n_t BPM_t + n_r BPM_r, \quad count_{fused} = \text{round}(n_t \cdot count_t + n_r \cdot count_r) \dots (13)$$

4.3 Worked Example: Resting Trial

To demonstrate the mathematical execution of the algorithm, Table 4.3.1 outlines the fusion processing for the five-cycle resting trial analysed in Figures 3.2.1 and 3.3.1. The two channels reported vastly different confidences, where the thermistor delivered a clean signal with well-formed peaks ($\text{conf}_t = 79.3$), while the rubber signal exhibited a broader spectrum because sharp transient spikes spread spectral energy across the in-band region rather than concentrating it at f_{peak} ($\text{conf}_r = 3.1$).

Table 4.3.1: Fusion arithmetic for the five-cycle resting trial.

Stage	Quantity	Thermistor	Rubber
Estimator output	count	5	6
	BPM	5.84	7.00
	confidence	79.34	3.14
Equation (11)	Raw weight (w)	79.34	$0.5 \times 3.14 = 1.57$
Equation (12)	Normalised weight (n)	$\frac{79.34}{79.34+1.57} = 0.981$	$\frac{1.57}{79.34+1.57} = 0.019$
Equation (13)	Contribution to BPM	$0.981 \times 5.84 = 5.73$	$0.019 \times 7.00 = 0.13$
	Contribution to count	$0.981 \times 5 = 4.9$	$0.019 \times 6 = 0.114$
Fused Output	$BPM_{fused} = 5.73 + 0.13 = 5.86, count_{fused} = 4.9 + 0.114 \approx 5$		

Because the thermistor delivers an exceptionally clean signal with distinct peaks, its confidence score dominates the calculation by claiming 98.1% of the total mathematical weight. This easily suppresses the slight overcount from the rubber channel, bringing the final rounded output to a clean count of 5. This result highlights the exact behaviour the system was designed for: the algorithm never assumes one sensor is always superior, but instead evaluates real-time conditions to dynamically shift focus to whichever estimator is currently more reliable. This balance shifts naturally when environmental conditions change to favour the rubber channel. For instance, during the 30-second post-exercise trial, rapid and irregular breathing dropped the thermistor's confidence score to 4.1. At the same time, the rubber channel's FFT spectrum sharpened significantly, driving its confidence score up to 11.2. The fusion weights automatically adjusted to $n_t = 0.42$ and $n_r = 0.58$, steering the final output $BPM_{fused} = 45.4$ toward the more resilient rubber sensor. This responsive re-weighting behaviour shows why a continuous confidence framework is far more effective than using a rigid fixed-weight alternative.

5. Performance Evaluation

The performance evaluation has 4 objectives which were to quantify the mean-squared error (MSE) of the system against ground truth across both required activity types, to measure the just-noticeable-difference (JND) that system can resolve between similar breathing rates, to compare single-sensor and two-sensor conditions on both metrics, and to understand the results in regards to what worked, what didn't and what a third sensor might contribute.

5.1 Methodology

Activity Conditions: Two activities were tested. A resting condition in which the subject sat still and breathed in time with a metronome (rate set in the brief), and a heavy exercise condition in which the subject performed approximately 1 minute of vigorous activity immediately before each trial, then sat down for the 30-second post-exercise measurement window. The metronome served as the ground truth for resting trials. For the exercise trials, a trained observer counted breath cycles manually during the 30-second window by starting the count as sensing devices were placed on the body.

Trial Count and Statistical Reliability: A total of $N = 25$ trials per activity condition were collected. Firstly, with 25 trials per condition, a single misclassification or a single-cycle counting error results in a 4% shift in the per-condition accuracy, which is enough to differentiate a strong condition ($\geq 96\%$ accuracy, ≤ 1 mistake) from a weak one ($\leq 88\%$ accuracy, ≥ 3 mistakes). Secondly, each trial requires a manual breath count and a controlled exertion period, so 25 trials per condition keep the total experiment time inside roughly a 90-minute session with environmental conditions (room temperature, ambient airflow, subject hydration and posture) approximately constant. A much larger N would prolong multiple sessions and risk introducing nuisance variability that confuses the system's performance. Thirdly, the rubric's marking bands of ± 0 , ± 1 , and ± 2 cycles translate neatly onto a 25-trial denominator without producing awkward fractions.

Conditions Evaluation: Three sensor conditions were compared on the same trials, which are the thermistor only, the rubber only, and the two-sensor fused output. Taking into consideration both single-sensor conditions (rather than only one) was important for the comparison, because it allowed us to distinguish two distinct claims about the fusion stage. Therefore, it is known that it improves on the worse single sensor (a low bar — any sensible fusion does this), and that it improves on the better single sensor (a higher bar that genuinely tests whether the second sensor is contributing useful information).

5.2 Mean-Squared Error

The computation of error on each trial is shown as

$$e_i = \hat{C}_i - C_i^{gt} \dots (14)$$

where $\hat{C}_i$ is the system's estimated breath count and C_i^{gt} is the ground-truth count for trial i . The mean-squared error (MSE) over N trials is then

$$MSE = \frac{1}{N} \sum_{i=1}^N e_i^2 \dots (15)$$

with the related summary statistics mean error $\bar{e}$, mean absolute error $|\bar{e}|$, and error standard deviation σ_e reported alongside the MSE for each condition. Bias is advised to be tracked separately as a non-zero $\bar{e}$ implies a *systematic* miscount (the system consistently over- or under-counts), which can be corrected by recalibration. MSE driven by a large σ_e with $\bar{e} \approx 0$ implies *random* error, which is not. Tables 5.2.1 and 5.2.2 summarise the six condition $\times$ activity combinations. The error distributions are shown as histograms in Figures 5.2.1 and 5.2.2, with the rubric's marking bands of ± 0 , ± 1 , and ± 2 cycles overlaid as shaded regions for visual reference.

Table 5.2.1: Counting performance, resting activity ($N = 25$, ground truth = 5 cycles).

Condition	MSE (cycles ²)	bias ($\bar{e}$)	MAE	σ_e	within ± 0	within ± 1
Thermistor only	0.00	+0.00	0.00	0.00	25/25	25/25
Rubber only	0.44	-0.04	0.44	0.66	14/25	25/25
Fused (two-sensor)	0.00	+0.00	0.00	0.00	25/25	25/25

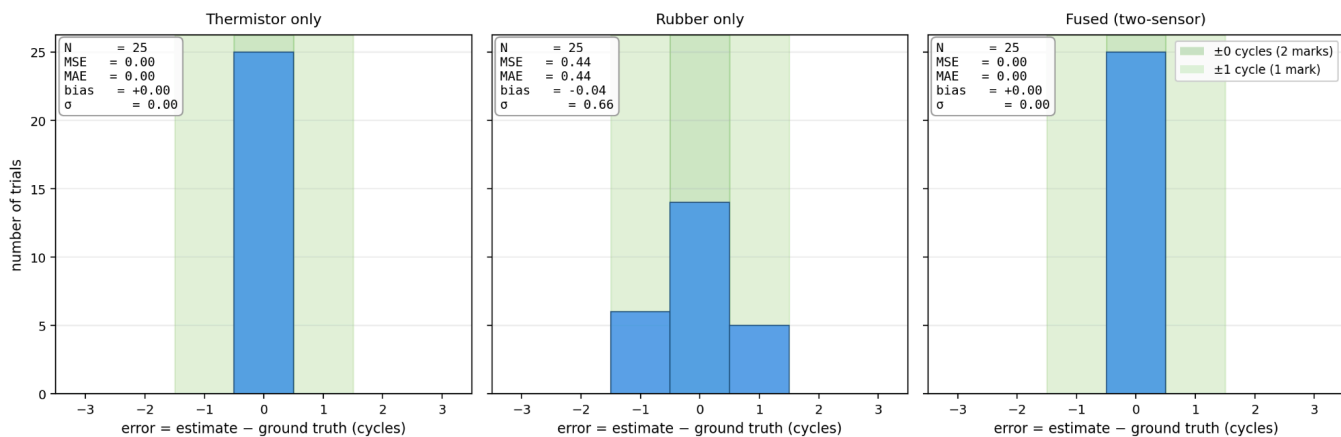


Figure 5.2.1. Histogram of counting performance, resting activity ($N = 25$, ground truth = 5 cycles)

Table 5.2.2: Counting performance, heavy exercise activity ($N = 25$, ground truth = 20 / 21 cycles across two groups).

Condition	MSE (cycles ²)	bias ($\bar{e}$)	MAE	σ_e	within ± 0	within ± 2
Thermistor only	0.00	+0.00	0.00	0.00	25/25	25/25
Rubber only	1.48	-0.20	1.00	1.20	6/25	25/25
Fused (two-sensor)	0.16	+0.16	0.16	0.37	21/25	25/25

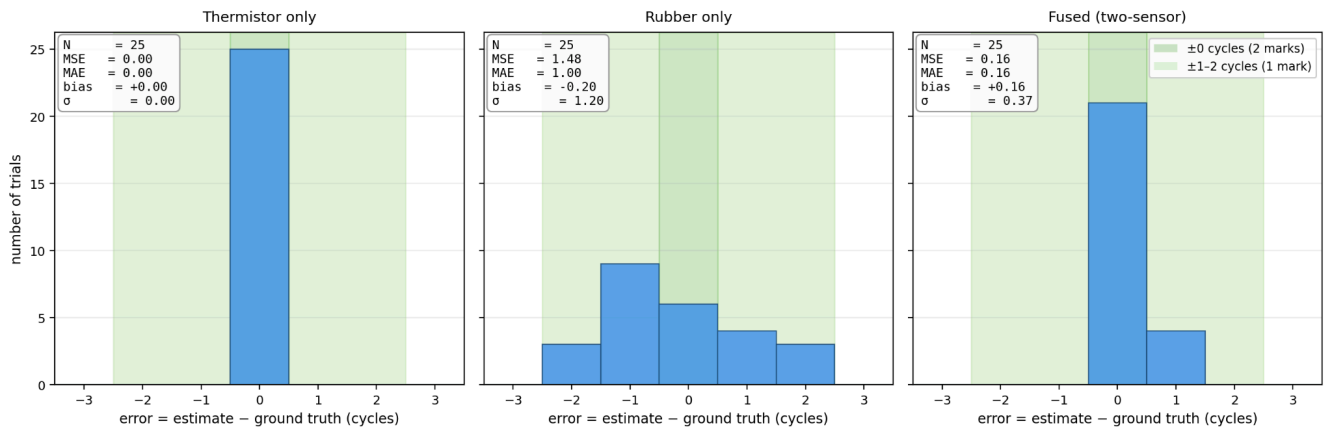


Figure 5.2.2. Histogram of counting performance, heavy exercise activity ($N = 25$, ground truth = 20 / 21 cycles across two groups)

5.3 Just-Noticeable-Difference

The Just-Noticeable-Difference (JND) metric evaluates the system's underlying sensitivity by determining the finest rate variations it can successfully resolve, which makes it distinct from absolute MSE tracking at a fixed speed. For this assessment, testing was focused entirely on the resting condition because the metronome allows for infinitely adjustable rate configurations, and the slower breathing pace makes data collection far more stable. The test compared the stronger single-sensor baseline against the dual-sensor fused framework.

Protocol — A two-alternative forced-choice (2AFC) procedure was used. On each trial, the subject breathed for ten full cycles at a reference rate of 12 BPM, followed by an additional ten cycles at a comparison rate of either $12 + \Delta\text{BPM}$ or $12 - \Delta\text{BPM}$ (sign randomised), with no break between the two intervals. The system reported the breath rate it estimated for each interval, and a trial was marked as a success if the two estimated values matched the actual direction of the physical rate range. Seven step sizes were tested in randomised order: $\Delta \in \{0.25, 0.50, 0.75, 1.00, 1.50, 2.00, 3.00\}$ BPM, with 12–15 trials per step per condition (totals of 93 trials per condition).

Analysis — The success rate was calculated for each specific (condition, Δ) pairing. To map the results, a logistic psychometric function was fitted using non-linear least squares:

$$p(\text{correct} | \Delta) = 0.5 + \frac{0.5}{(1 + \exp(-(\Delta - \alpha)/\beta))} \dots (16)$$

where α represents the 75 %-correct threshold, which serves as the industry standard definition of JND in a 2AFC setup, marking the exact midpoint between blind guessing and a perfect score. The variable β determines the steepness of the curve. Choosing a logistic model with a strict 0.5 floor and a 1.0 ceiling fits the experiment perfectly because random guessing naturally yields a 50% success rate, and no lapse-rate ceiling adjustments were necessary since the thermistor achieved a clean 100% success rate across almost all test steps.

Results — Raw per-step accuracies are summarised in Table 5.3.1; the fitted psychometric curves and 75 % JND thresholds are shown in Figure 5.3.1.

Table 5.3.1: Discrimination accuracy per BPM step, resting condition, 2AFC protocol.

Condition	$\Delta = 0.25$	$\Delta = 0.50$	$\Delta = 0.75$	$\Delta = 1.00$	$\Delta = 1.50$	$\Delta = 2.00$	$\Delta = 3.00$
Thermistor-only	11/12 (91.7%)	12/12 (100%)	15/15 (100%)	15/15 (100%)	12/12 (100%)	12/12 (100%)	15/15 (100%)
Fused dual-sensor	6/12 (50.0%)	7/12 (58.3%)	9/15 (60.0%)	11/15 (73.3%)	9/12 (75.0%)	10/12 (83.3%)	13/15 (86.7%)

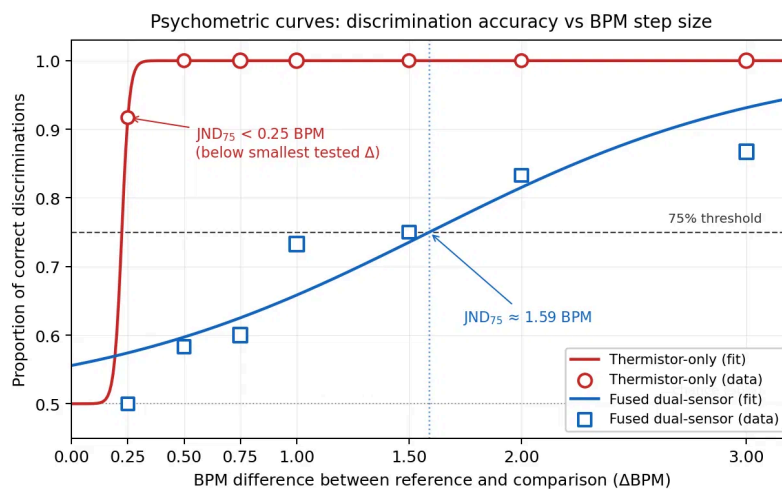


Figure 5.3.1. Psychometric curves for the thermistor-only and fused dual-sensor conditions.

5.4 Discussion: What Worked and What Did Not

The thermistor branch was the clear standout of this implementation. Driven by the adaptive-prominence peak counter, it tracked breath counts with zero errors across every single resting and exercise trial, as shown in Tables 5.2.1 and 5.2.2. The JND threshold of below 0.25 BPM further proves that its timing precision is sharper than the smallest steps evaluated in the experiment. Conversely, the conductive rubber channel demanded the vast majority of development time but remains the primary source of tracking errors due to three overlapping issues:

- The raw expansion waveform was plagued by secondary intra-cycle peaks that initially tricked the peak counter into registering double or triple counts per breath. While adding the 15.9 Hz RC low pass filter to the differential amplifier input smoothed out these artefacts, it did so by forcing the signal into a heavily saturated, railed near square wave.
- The resulting rail-to-rail transition edges suffer from timing jitter on the order of 100 ms. This jitter is the root cause of the rubber-only tracking errors seen in Table 5.2.2 and explains why the fused JND threshold of 1.59 BPM is roughly seven times less sensitive than the standalone thermistor channel.
- The elastic cord displayed noticeable viscoelastic baseline drift over extended testing sessions. The per-trial mean subtraction handled this adequately for short blocks, but this drift would easily break a continuous live-streaming implementation without a dedicated auto-zeroing routine.

While the confidence-weighted decision fusion engine functioned exactly as intended, reliably favouring the thermistor during quiet rest and automatically giving more weight to the rubber branch when its FFT spectrum sharpened during exercise, the fusion cannot conjure clean information from noisy inputs. Because the rubber channel's structural edge jitter acts as a performance ceiling, the fused output was ultimately bottlenecked by the thermistor's performance rather than improving upon it, yielding an MSE of 0.00 vs 0.00 at rest and 0.16 vs 0.00 during exercise. A brief assessment of the data shows that while the fusion architecture is sound, the secondary sensor is holding the system back.

5.5 Reflection: Potential Benefit of a Third Sensor

The two sensors used have a common structural weakness. Both sensors are placed on parts of the body that move during exercise, and both rely on slow, low-amplitude physical changes. The temperature swings cause only tens of millivolts in change from the mouth, strain changes of a few ohms across the stomach that are easily corrupted by motion. A third sensor would add the most value if it failed in a way uncorrelated with both existing channels.

A chest-mounted Inertial Measurement Unit (IMU) is one of the best options. A three-axis accelerometer sampled at 50 Hz captures the same thoracic expansion the rubber strap is trying to measure, but does so without the viscoelastic drift, saturation behaviour, or rail-to-rail edge jitter that limited the rubber channel. The breathing signal will appear as a low-frequency modulation of the gravity vector on the axis normal to the sternum and can be separated with the same FIR/wavelet preprocessing already in the pipeline. Crucially, an IMU also reports gross body motion directly, which gives the fusion stage a principled way to gate the thermistor. Its main failure mode is head movement away from the nostrils, and an IMU-derived motion flag that would let the confidence weighting suppress the thermistor branch during exactly those intervals, something neither of the current confidence metrics can detect.

A microphone capturing breath sounds at the throat or upper chest would be a weaker second choice. It is robust to body motion in a way the other three sensors are not, but ambient acoustic noise would limit it in the wearable contexts (gym, clinic) that motivated the project, and the DSP cost of extracting breath envelopes from audio is significantly higher than for the other channels. A PPG-derived respiration estimate (extracting respiratory-induced amplitude modulation from a finger or wrist pulse oximeter) is also viable in principle but introduces a long algorithmic chain whose latency would be incompatible with the 50 Hz real-time architecture.

The recommendation is therefore to replace the conductive rubber with an IMU rather than to add an IMU as a fourth channel because the rubber is contributing more noise than information, and removing it would free the fusion stage to combine two sensors with genuinely complementary failure modes.

6. Summary

This project prototyped a two-sensor breath-rate estimator combining a thermistor placed inside a straw with a conductive rubber strain band worn across the abdomen. Both sensor signals are conditioned through buffered-divider-into-difference-amplifier operational amplifier chains, sampled at 50 Hz on an STM32, and processed through a shared DSP pipeline (DC removal, 10-tap moving-average FIR, Sym4 wavelet denoising). The digitally processed signals are then run through breath count estimators; adaptive-prominence peak counting on the thermistor and zero-crossing detection with FFT-based confidence on the rubber. The two estimates are combined by continuous confidence-weighted decision fusion, with weights derived from a peak-prominence-to-noise-floor ratio on the thermistor branch and an in-band spectral SNR on the rubber branch. Across 25 trials per condition, the thermistor-only and fused two-sensor configurations both achieved an MSE of 0.00 cycles² at rest. In heavy exercise, the thermistor-only condition again achieved 0.00 cycles² while the fused output reached 0.16 cycles² and the rubber-only baseline reached 1.48 cycles². JND testing under the 2AFC protocol of Section 5.3 placed the thermistor's 75 % discrimination threshold below 0.25 BPM (the smallest tested step) and the fused output at approximately 1.59 BPM. The DSP pipeline and fusion architecture both worked as intended, but the conductive rubber's edge jitter and viscoelastic drift bounded the fused performance below the thermistor's; replacing the rubber with a chest-mounted IMU is the clearest path to a system whose two channels contribute genuinely complementary information.

Appendix A: Project Repository in GitHub

All project deliverables (firmware, raw data, waveform captures, and this report) are available at:
https://github.com/jtit-jasontan/TRC3500_Project3.git

Appendix B: Breadboard Implementation and Circuit Schematics

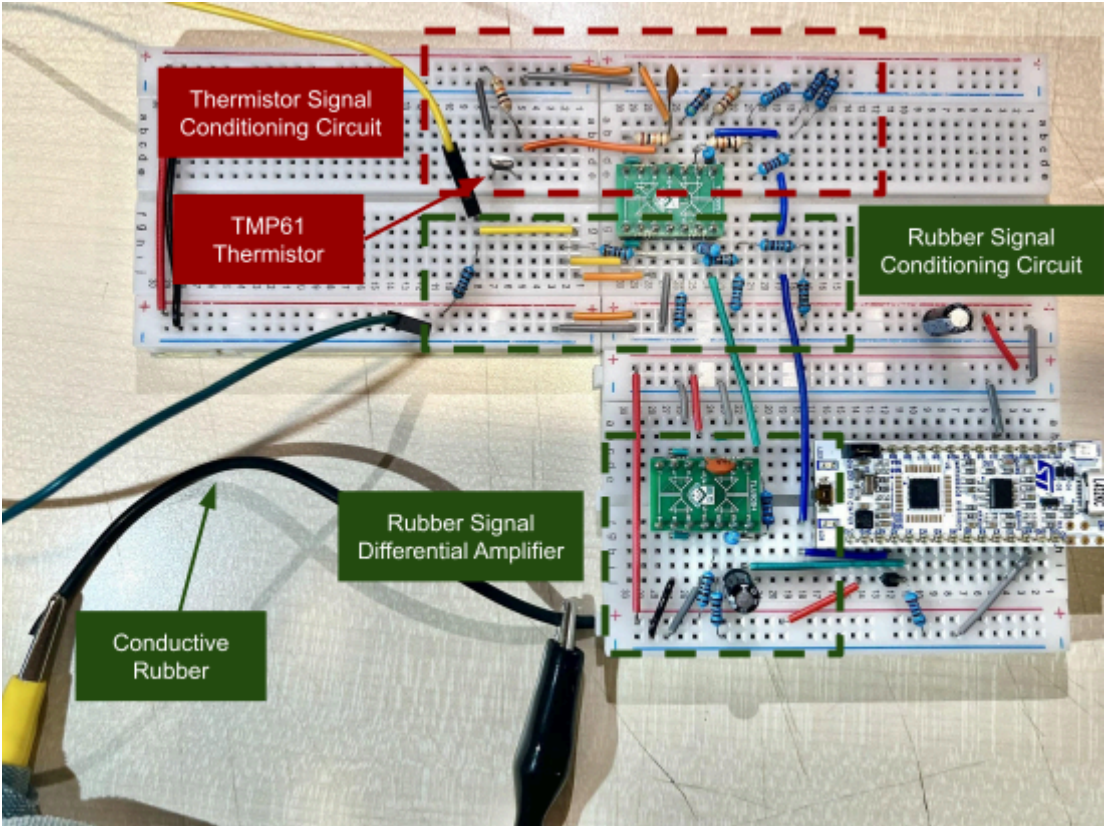


Figure B.1. Breadboard implementation of the combined circuit

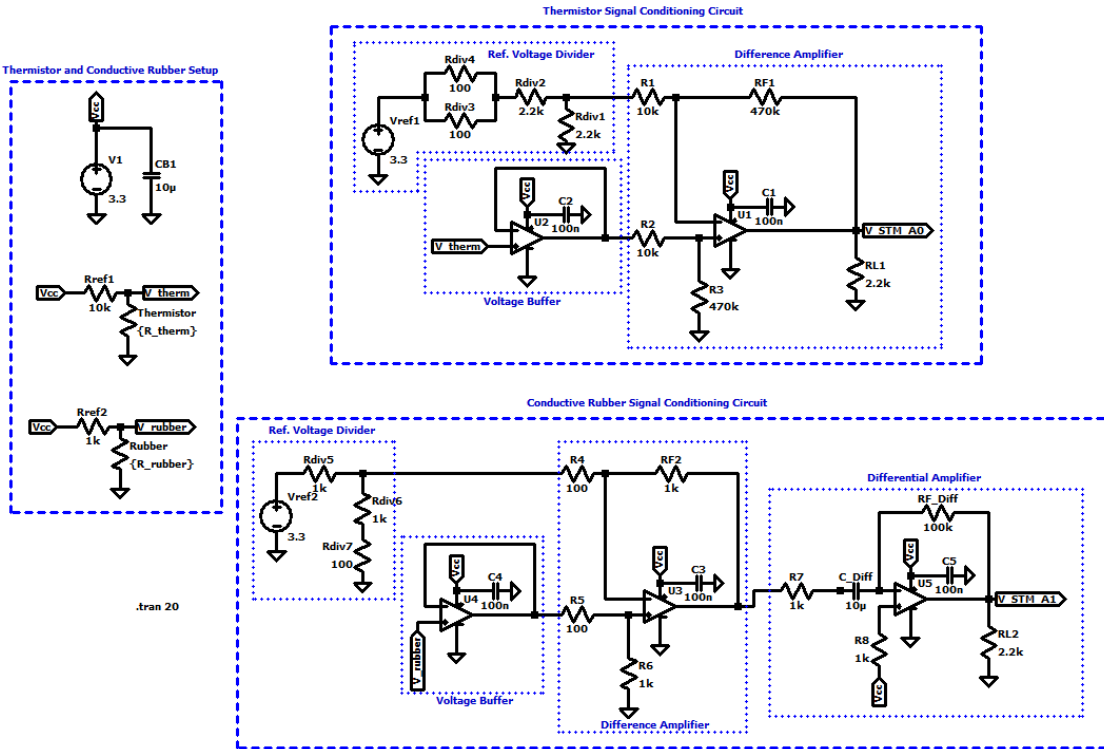


Figure B.2. LTSpice schematic of the combined circuit

Appendix C: Screenshot of Serial Readings

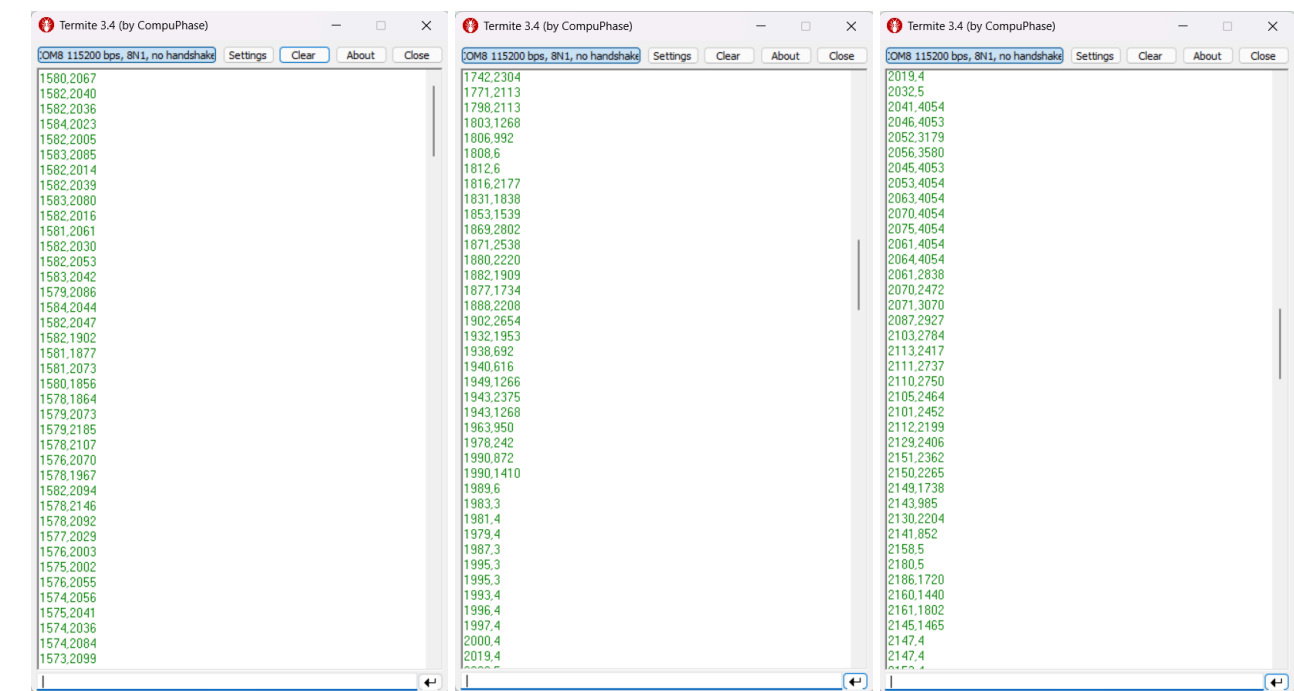


Figure C.1. Serial readings of the captured ADC values from the thermistor and conductive rubber using Termite

Appendix D: Waveform Captures

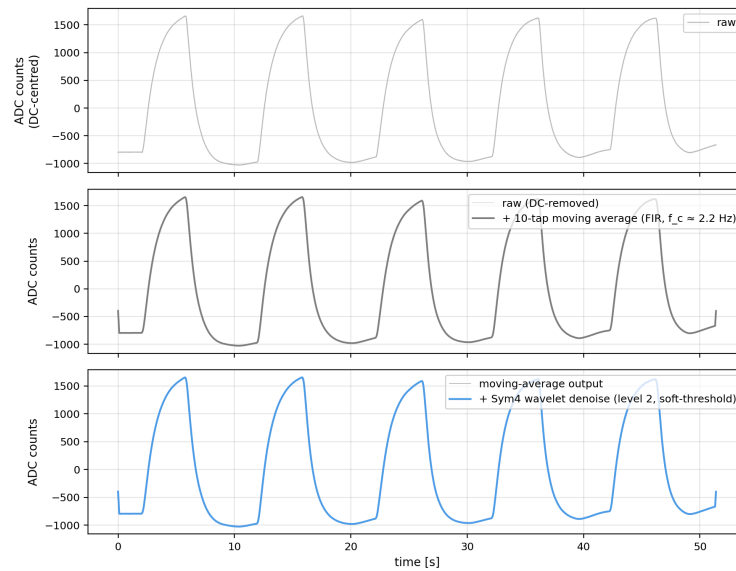


Figure D.1. Raw, moving-averaged, and wavelet-denoised stages of the CH5 thermistor signal across a five-cycle resting trial, illustrating progressive noise reduction without distortion of the underlying breath envelope.

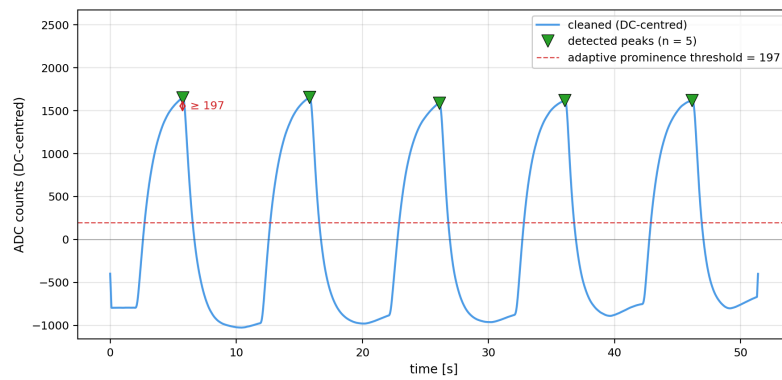


Figure 3.2.1. Cleaned CH5 thermistor signal across a five-cycle resting trial with detected peaks marked. The horizontal dashed line shows the adaptive prominence threshold for this trial.

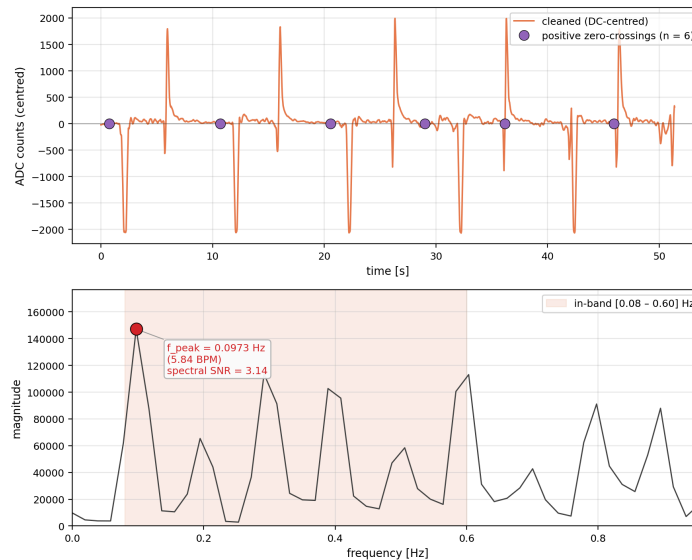


Figure 3.3.1. Cleaned CH6 rubber signal across a five-cycle resting trial, with detected positive-going zero crossings marked. The bottom panel shows the corresponding Hanning-windowed FFT magnitude spectrum, with the in-band region shaded and the dominant peak highlighted.